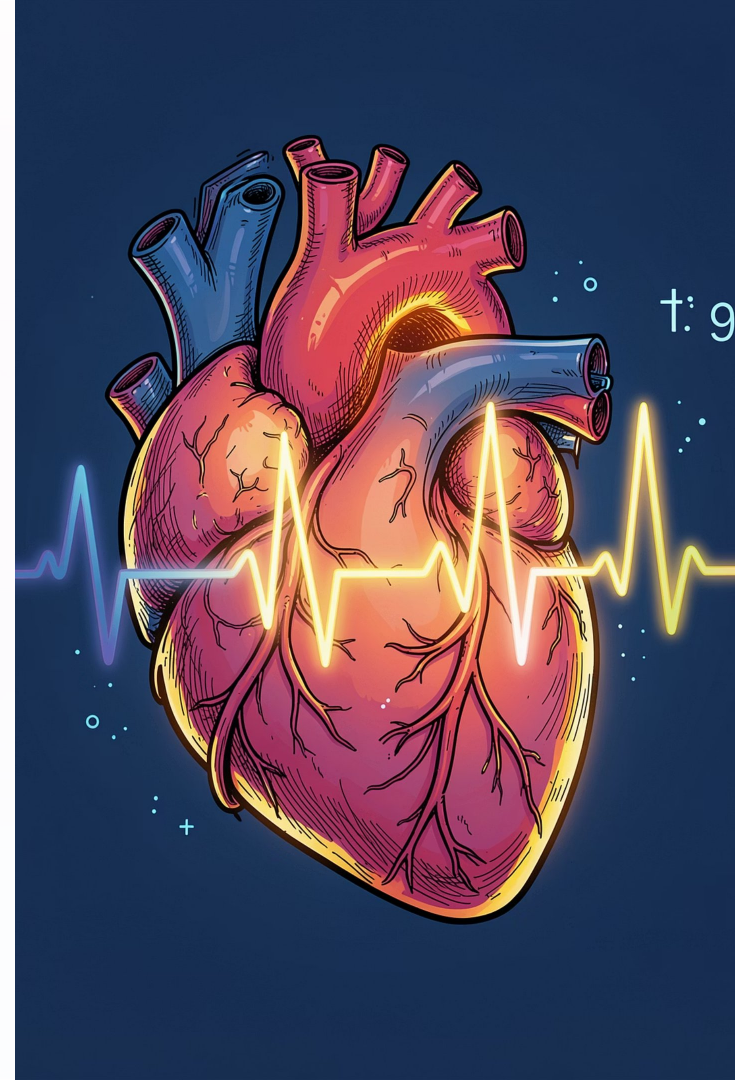
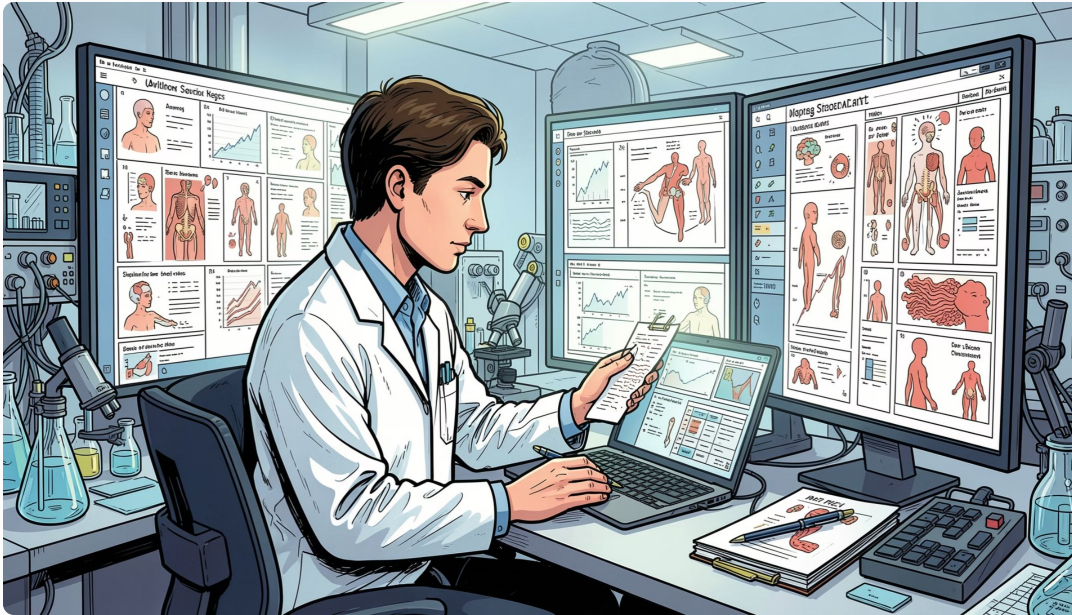


Final Clustering Project: Heart Disease Dataset

Unsupervised Learning — K-means Clustering



Introduction & Objective



Approach

Applying **Unsupervised Learning** via K-means Clustering to group patients based solely on biomedical measurements — with no disease labels used during training

Methodology

A structured data science pipeline: data loading → cleaning → feature transformation → optimal cluster estimation → model evaluation

Dataset Verification & De-duplication

1,025

Initial Records

Patient profiles with 14 biomedical features

0

Missing Values

Thoroughly checked across all columns

723

Duplicates Removed

Critical step to prevent algorithmic bias

302

Final Dataset

Unique, high-quality patient profiles

 Removing 723 duplicate rows was essential — without this step, the clustering algorithm would have been biased toward overrepresented records

Preparing for Unsupervised Learning

Target Column Dropped

The disease diagnosis column was completely removed from the training set to strictly follow the unsupervised paradigm

Target Saved for Validation

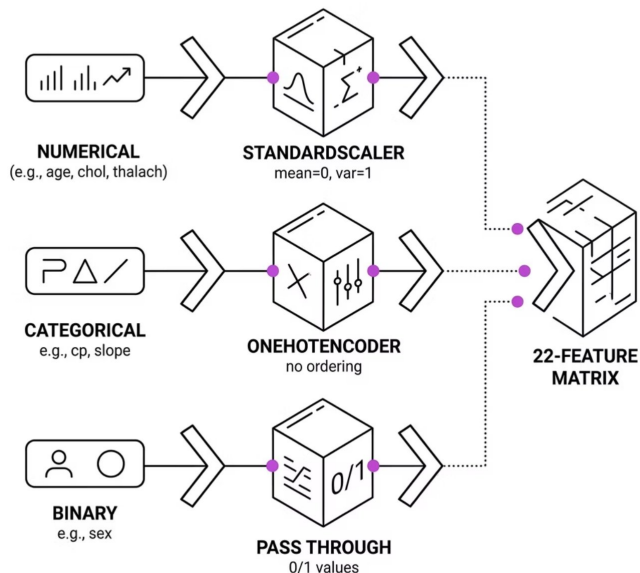
The label was stored separately and used *only* for post-hoc validation and cluster interpretation

Final Feature Matrix

X consists of 13 medical attributes — age, cholesterol, maximum heart rate and more — ready for clustering

Data Preprocessing Pipeline

.A ColumnTransformer ensured fair distance calculations in K-means by treating each feature type appropriately



Result

The pipeline transformed 13 raw attributes into an optimized matrix of **features** — properly scaled and encoded — ready for K-means clustering

- (1 = Numerical: StandardScaler (mean = 0, variance = 1))
- (0 = Categorical: OneHotEncoder (no arbitrary ordering))
- (1 / 0 Binary: Passed through as is)

Evaluating K Values

Two Metrics, One Decision

K values from 2 to 10 were evaluated using **Inertia** (Elbow Method) and **.Silhouette Score**



K=2 selected — highest Silhouette score and
.clinically meaningful: Healthy vs. Heart Disease

2 = K

1869.45 :Inertia

✓ **0.177** :Silhouette

3 = K

1701.81 :Inertia

0.159 :Silhouette

4 = K

1591.32 :Inertia

0.120 :Silhouette

MODEL TRAINING

K-means Model Training

Configuration

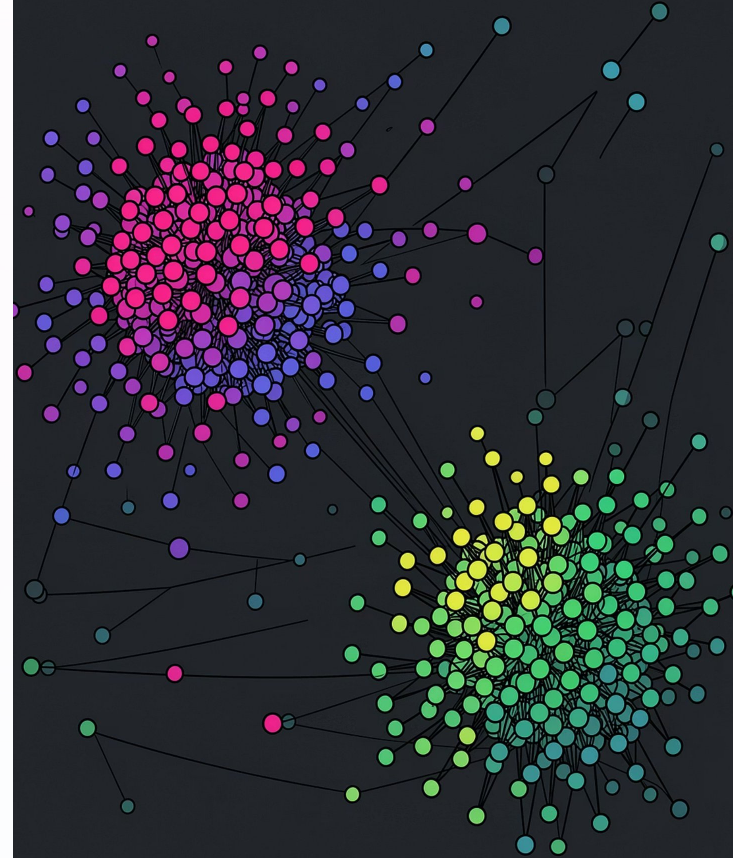
`-n_clusters=2` with `init='kmeans++'` for optimal centroid initialization and faster convergence

Segmentation

All 302 patients segmented into two distinct groups based on 22 patterns across the preprocessed features

Best Performer

This configuration achieved the mathematically superior partition compared to all higher values of K tested



Cluster Profiles & Medical Interpretation

Cluster 0 (Higher Risk)

Cluster 1 (Lower Risk)

Clinical Alignment

Cross-tabulation with hidden labels revealed the two clusters correspond strongly to **heart disease vs. healthy patients** — without .ever seeing the diagnosis

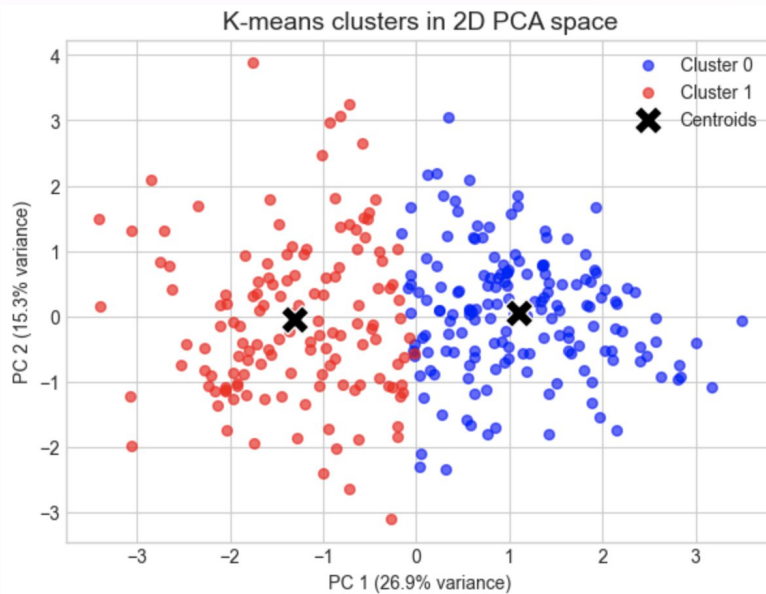
Key Drivers

thalach — maximum heart rate achieved •

oldpeak — ST depression induced by exercise •

cp — chest pain type •

exang — exercise-induced angina •



Silhouette score ~0.17 is positive but low — expected in medical data, where patient profiles exist on a continuum rather than forming .perfectly isolated groups



CONCLUSION

Project Takeaways

1

De-duplication Was Critical

Removing 723 duplicate rows prevented false patterns and ensured an unbiased model

2

Mixed-Type Preprocessing Works

-Combining `StandardScaler` and `OneHotEncoder` allowed K-means to handle diverse medical data fairly

3

Unsupervised Learning Has Real Value

Without seeing diagnosis labels, K-means discovered meaningful biological groups — demonstrating its potential for preliminary patient stratification in clinical settings

Thank You

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